Peter Juhasz — Researcher

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ResearchGate • LinkedIn • GitHub

Summary

I am a PhD Candidate in Mathematics at Aarhus University, supervised by Christian Hirsch. My doctoral research combines stochastic processes and topological data analysis to create novel models for complex networks. Prior to my PhD, I worked as a deep learning researcher in automated driving at Bosch Group Hungary, where I developed algorithms for vehicle perception, prediction and decision-making. Additionally, I led the design and implementation of scalable software solutions as a software architect in the same field.

Education

PhD in Mathematics
Aarhus University, Denmark, Stochastic Point Processes and Topological Data Analysis *2022–*
Supervisors: Dr. Christian Hirsch, Dr. Claudia Strauch
Thesis: Point Process Based Models of Evolving Higher-Order Networks

Master of Business Administration with Highest Honors
Budapest University of Technology and Economics, Hungary, Specialized in Finance *2019–2021*
Supervisor: Dr. Roland Molontay
Thesis: Prediction of Popularity of Memes using Machine Learning Methods

Master of Science in Physics
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics *2013–2016*
Supervisor: Dr. Gabor Vattay
Thesis: Examination of Spatially Embedded Complex Networks
Erasmus+ Scholarship: Delft University of Technology (Applied Physics)

Bachelor of Science in Physics
Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics *2010–2013*
Supervisor: Dr. Szabolcs Beleznai
Thesis: Finite Element Method Plasma Simulation of Nitrogen Contaminated Ceramic Metal Halide Lamps

Professional Experience

Research Assistant in Spatial Stochastics and Topological Data Analysis
Department of Mathematics, Aarhus University, Denmark *2022–*
- Published 4 papers in stochastic models for higher-order networks
- Implemented topological data analysis algorithms on simulated spatial point processes
- Spent 3 months at TU Braunschweig supervised by
  Dr. Benedikt Jahnel (Weierstrass Institute for Applied Analysis and Stochastics, Berlin)
- Taught 5 courses in mathematics and computer science

Deep Learning Researcher in Automated Driving
Department of Automated Driving, Bosch Group Hungary *2021–2022*
- Patented a Gaussian regression based trajectory prediction model for automated driving
  Analyzed more than 20 000 hours of measurements of passenger car driving
  Enhanced prediction accuracy by a factor of 3 and the uncertainty estimation by a factor of 10
- Object detection and classification with ultrasonic sensors:
  Designed several features to represent objects and free space
  Implemented a U-Net neural network to detect and classify objects
- Real time object tracking with temporal convolutional neural networks:
  Improved a feature pyramid network to track objects in real time

Software Architect in Automated Driving
Department of Automated Driving, Bosch Group Hungary *2018–2021*
- Lead a team of 11 in environment prediction as a technical lead
- Created a concept for automated ground-truth generation for data-driven machine learning development
- Coordinated 2 teams in the integration of driver monitoring camera for hands-free driving

Software Engineer for Mobile Networks
Business Unit IT & Cloud Products, Ericsson Telecommunications Hungary *2016–2018*
- Analyzed response times of distributed database servers for optimal load balancing
- Coordinated a team of 8 as a deputy of the product owner
- Implemented performance critical algorithms for Smart Services Routers

Publications, Patents

C. Hirsch, B. Jahnel and **P. Juhasz**. *2025*
Functional limit theorems for edge counts in dynamic random connection hypergraphs,
arXiv:2507.16270, 2025, DOI 10.48550/arXiv.2507.16270.

C. Hirsch and **P. Juhasz**. *2025*
On the topology of higher-order age-dependent random connection models,
Methodol. Comput. Appl. Probab. **27** (2025), no. 2, DOI 10.1007/s11009-025-10173-7.

C. Hirsch, B. Jahnel, S. K. Jhawar and **P. Juhasz**. *2025*
Poisson approximation of fixed-degree nodes in weighted random connection models,
Stochastic Process. Appl. **183** (2025), 104593, DOI 10.1016/j.spa.2025.104593.

M. Brun, C. Hirsch, **P. Juhasz** and M. Otto. *2024*
Random connection hypergraphs,
arXiv:2407.16334, 2024, DOI 10.48550/arXiv.2407.16334.

K. Barnes, **P. Juhasz**, M. Nagy and R. Molontay. *2024*
Topicality boosts popularity: a comparative analysis of NYT articles and Reddit memes,
Soc. Netw. Anal. Min. **14** (2024), no. 119, DOI 10.1007/s13278-024-01272-3.

P. Juhasz. *2021*
Information propagation in stochastic networks,
Phys. A **577** (2021), 126070, DOI 10.1016/j.physa.2021.126070.

P. Juhasz. *2021*
Method and device for predicting the trajectory of a traffic participant, and sensor system,
CNIPA: CN113988353A, EPO: EP3916697A1, JPO: JP2021190119A, USPTO: US2021366274A1, 2021.

P. Juhasz, J. Steger, D. Kondor and G. Vattay. *2018*
A Bayesian approach to identify Bitcoin users,
PLoS One **13** (2018), e0207000, DOI 10.1371/journal.pone.0207000.

P. Juhasz, Sz. Beleznai and I. Maros. *Finite element method plasma simulation* *2014*
Finite element method plasma simulation of nitrogen contaminated metal halide lamps
Comsol Conference Proceedings, 2014.

Awards, Grants

Individual Travel Grant
European Mathematical Society *2025*
500 EUR – Talk "Random Connection Hypergraphs" at GPSD 2025 conference

Academic Mobility Grant for International Exchange
Aarhus University *2024*
20 000 DKK – Evolving Stochastic Network Models

PhD Fellowship Grant (DDSA-PhD-2022-008)
Danish Data Science Academy *2022–2025*
1 890 000 DKK – Topological Data Analysis Based Models of Evolving Higher-Order Networks

Excellence Award for Outstanding Thesis and Diploma Work
Hungarian National Bank — *2021*
350 000 HUF – Digitalization, Artificial Intelligence, and the Age of Data – What Makes a Meme Viral?

Conferences, Events

Talks

Scaling Limits in Spatial Birth-Death Systems with Long-Range Interactions
Stochastic Interacting Particle Systems and Random Matrices — *2025*

Random Connection Hypergraphs
German Probability and Statistics Days 2025 — *2025*

Stochastic Random Connection Models
Probability Seminar — *2024*

Topological Data Analysis Based Models of Evolving Higher-Order Networks
Qualifying Exam — *2024*

Adaptive Trajectory Prediction Based on a Gaussian Regression Model
Bosch Group: Bosch Innovation Day — *2020*

Probabilistic Localization of Bitcoin Users
Hungarian Academy of Sciences: Statistical Physics Conference — *2016*

Posters

Functional Stable Limit in Random Connection Hypergraphs
Topological data analysis in stochastic geometry and image processing — *2025*

Topological Data Analysis of Higher-Order Networks
Aarhus University Network on Artificial Biology Symposium — *2025*

Simplex Count Distribution in Random Connection Hypergraphs
Particle Systems in Random Environments — *2024*

Distribution of Betti Numbers in Age-Dependent Random Connection Models
DDSA Annual Conference — *2024*

Topological Data Analysis of Higher-Order Networks
Danish–Swedish Summer School on TDA and Spatial Statistics — *2023*

Dissecting Distributed Database Systems For Telecom Applications
Ericsson Telecommunications Hungary: Ericsson University Day — *2016*

Finite Element Method Plasma Simulation of Ceramic Metal Halide Lamps
Comsol Multiphysics Conference — *2014*

Programming Skills

Python	● ● ● ● ●
Git	● ● ● ● ○
C++	● ● ● ● ○

Languages

Hungarian	● ● ● ● ●
English	● ● ● ● ○
German	● ● ○ ○ ○

Teaching

Advanced Probability Theory
Aarhus University, Course Code: F25.550171U020 *2025*
10 ECTS Exercise Class, Spring Semester

Data Project
Aarhus University, Course Code: F24.550201U004 *2024*
10 ECTS Computer Project Class, Spring Semester

Linear Transformations
Aarhus University, Course Code: E23.550171U010 *2023*
10 ECTS Exercise Class, Autumn Semester

Ordinary Differential Equations
Aarhus University, Course Code: F23.550141U011 *2023*
5 ECTS Exercise Class, Spring Semester

Advanced Calculus for Engineers
Aarhus University, Course Code: F23.280191U021 *2023*
10 ECTS Exercise Class, Spring Semester